MinVIS Strikes Back: Robust Training-Free Tracking via Multi-Frame Query Aggregation

Danial Hamdi, Fardin Ayar & Mahdi Javanmardi
Department of Computer Engineering
Amirkabir University of Technology
Tehran, Iran
{danialz,fardin,mjavan}@aut.ac.ir

Abstract

We revisit MinVIS, an online video instance segmentation (VIS) framework noted for achieving competitive tracking performance without video-based training. MinVIS first trains a query-based model on independent image frames; during inference, it tracks instances by matching their corresponding query embeddings between consecutive frames. While this approach is effective, its reliance on single-frame history can be brittle, where one suboptimal match can derail an entire object track. To quantify this fragility, we establish a performance upper bound by formulating an ideal matching algorithm via integer linear programming, revealing a significant gap (9.5 AP on YTVIS-19 Yang et al. (2019)) between the standard greedy approach and the optimal solution. Motivated by this, we introduce **MinVIS-W**, a robust tracking method based on multi-frame query aggregation. Our approach stabilizes tracking by computing a weighted similarity score across a temporal window of query embeddings, leveraging historical context to mitigate noise from frame-to-frame comparisons. This simple post-processing enhancement boosts performance on standard VIS benchmarks, achieving a **+3.21 AP gain** on the YouTube-VIS 2019 validation set, without requiring *any* retraining or architectural changes. The code is available at `https://github.com/da-nial/MinVIS-W`.

1 Introduction

Video Instance Segmentation (VIS) is a fundamental computer vision task that involves the simultaneous detection, segmentation, and tracking of object instances in video sequences Yang et al. (2019). While detection localizes instances and segmentation delineates their precise boundaries within each frame, tracking is the critical component that maintains instance identities over time. This ensures that an object is consistently recognized as it moves, interacts with other objects, or undergoes appearance changes.

VIS methods are often categorized into offline and online approaches. Offline, or per-clip, methods process an entire video or long clips at once to produce spatiotemporal masks, often achieving high accuracy at the cost of significant memory and computational requirements Cheng et al. (2021); Wang et al. (2021); Wu et al. (2021). In contrast, online, or per-frame, methods first process each frame with an image segmentation model and then associate instances between frames in a post-processing step to conduct tracking. This offers greater efficiency and suitability for real-time applications Yang et al. (2021); Huang et al. (2022).

A key driver of recent progress has been the advent of query-based instance segmentation models like Mask2Former Cheng et al. (2022). These architectures utilize a set of learnable vectors, known as **queries**, that act as placeholders for potential objects. Through a Transformer-based attention mechanism, these queries interact with image features to specialize, with each query ultimately becoming responsible for predicting the mask and class of a single object instance. The final output is a set of rich, high-dimensional query embeddings, each encapsulating the identity of one object in the frame.

MinVIS is a prominent online framework that leverages this architecture, demonstrating that a model trained exclusively on images can achieve competitive performance without any video-based training Huang et al. (2022). The framework operates in two distinct stages:

1. **Image-Level Segmentation**: During training, for each input frame $X \in \mathbb{R}^{H \times W}$, the query-based model produces a set of N query embeddings, $Q = \{q_i\}_{i=1}^{N}$, where $q_i \in \mathbb{R}^C$. Each query q_i is then passed to dedicated heads to predict a segmentation mask $m_i \in \{0, 1\}^{H \times W}$ and a class probability distribution $p_i \in \mathbb{R}^K$ over K classes.
2. **Training-Free Tracking**: During inference, this trained model is applied to each frame of a video. To track instances between two consecutive frames, t and $t+1$, MinVIS computes a cosine similarity matrix S between their respective query embeddings, Q^t and Q^{t+1}. The entry $S_{ij} = \cos(q_i^t, q_j^{t+1})$ represents the similarity between query i from frame t and query j from frame $t + 1$. The Hungarian algorithm is then used to find an optimal bipartite matching that maximizes the total similarity, thereby associating instances across frames.

A key innovation in MinVIS is delegating the matching task to the high-dimensional query embedding space. This makes the tracking resilient to occlusions and appearance changes, as it does not rely on fragile 2D mask-based heuristics like Intersection-over-Union (IoU). However, this frame-to-frame matching strategy, while effective, can be brittle. By design, the model generates far more queries than actual objects ($N \gg L$), meaning most queries are *null* and do not correspond to any instance (Figure 2). The Hungarian algorithm seeks a globally optimal assignment for all N queries, which can lead it to sacrifice the best match for an important object query in favor of a slightly better total score across all pairs, including the numerous null queries (Figure 3). A single such error can derail an entire track as the mistake propagates through subsequent frames (Figure A.1).

This reveals a potential gap between the greedy, frame-by-frame matching and what could be achieved with perfect foresight. In this paper, we first quantify this performance gap by formulating an **ideal matching oracle**. By using integer linear programming with access to ground truth annotations, we calculate an empirical upper bound for tracking performance. This analysis reveals the significant potential for improvement by addressing matching errors. Motivated by this finding, we propose **MinVIS-W**, a simple yet effective enhancement based on **multi-frame query aggregation**. Instead of relying solely on the previous frame, our method calculates an aggregated similarity score from a temporal window of past query embeddings. This leverages historical context to stabilize matching, making the tracking process more robust to noisy similarity scores and preventing catastrophic failures. Our approach is a training-free post-processing step that significantly improves tracking performance, which we detail in the following sections.

2 Related Works

Modern approaches to Video Instance Segmentation (VIS) have been heavily influenced by the success of Transformers in object detection Carion et al. (2020) and segmentation Cheng et al. (2022). Most online VIS methods follow a *tracking-by-detection* paradigm: an image instance segmentation model is applied to each frame, and a subsequent tracking mechanism associates the detected instances over time Wu et al. (2022). While early object trackers often used training-free association logic Wojke et al. (2017), the recent trend has shifted towards learnable tracking heads that are explicitly trained with video data to produce discriminative embeddings for robust association.

The pioneering work of MinVIS Huang et al. (2022) challenged this trend by demonstrating that the object queries from a transformer-based *image* segmentation model are already sufficiently discriminative for tracking. It achieved state-of-the-art performance with a simple query-matching algorithm, entirely avoiding video-based training for the tracking component. This *video-free* tracking philosophy, however, has been less explored in subsequent works.

Instead, many recent methods have focused on enhancing query-based tracking through video-level supervision. Techniques such as **contrastive learning** have been used to explicitly push queries of the same instance closer together across frames Wu et al. (2022). Another popular strategy is **query propagation**, where queries from a previous frame are refined and ”propagated” as initializers for the next frame’s prediction He et al. (2022); Li et al. (2023).

While these methods have advanced the state of the art, they typically require complex video-based training pipelines. The fundamental premise of MinVIS—that robust tracking can emerge from a purely image-trained model—remains a powerful but less-explored concept. Our work follows this direction, focusing not on adding new learnable components but on improving the robustness of the original training-free matching logic by better leveraging temporal information.

3 METHOD

Our methodology is designed to first diagnose the weaknesses in MinVIS's tracking, then propose a targeted, low-overhead solution. We begin by establishing a performance upper bound using an Integer Linear Programming (ILP) formulation, which we call an ideal **matching oracle**. We then introduce our qualitative analysis tool, **TrackLens**, to visually inspect failure modes. Motivated by these findings, we introduce our primary contribution: **MinVIS-W**, a multi-frame query aggregation method that enhances tracking robustness.

3.1 QUANTIFYING THE PERFORMANCE GAP: AN IDEAL MATCHING ORACLE

The standard tracking in MinVIS uses a greedy, frame-to-frame approach. To isolate the performance lost to this matching strategy (as opposed to suboptimal detection or segmentation), we formulate an **ideal matching oracle** using Integer Linear Programming (ILP). This oracle has access to the ground truth annotations and finds the globally optimal assignment of predicted queries to ground truth object tracks over an entire video, maximizing segmentation quality while respecting the constraints of the task.

A critical aspect of VIS evaluation is that a predicted track is assigned a single class label for its entire duration. This is typically determined by averaging the class logits of all query predictions assigned to the track and taking the class with the highest average score. Our formulation must therefore not only find matches with high segmentation quality but also ensure that the resulting tracks are assigned the correct class label.

Problem Formulation. Given a video segment of T frames, with M ground truth object tracks and N queries predicted per frame, we define a binary decision variable $x_{m,t,n}$ as:

$$x_{m,t,n} = \begin{cases} 1 & \text{if query } n \text{ in frame } t \text{ is assigned to object track } m \\ 0 & \text{otherwise} \end{cases}$$

The objective is to maximize the total Intersection-over-Union (IoU) over all assignments:

$$\text{maximize} \sum_{m=1}^{M} \sum_{t=1}^{T} \sum_{n=1}^{N} \text{IoU}(m,t,n) \cdot x_{m,t,n}$$

where $\text{IoU}(m,t,n)$ is the IoU between the ground truth mask for object m at frame t and the mask predicted by query n at frame t.

Constraints. This optimization is subject to several constraints:

1. **Unique Assignment per Object-Frame:** Each ground truth object must be assigned exactly one query in every frame it appears in.

$$\forall m, t: \quad \sum_{n=1}^{N} x_{m,t,n} = 1$$

2. **Unique Usage per Query-Frame:** Each query can be assigned to at most one ground truth object in any given frame.

$$\forall t, n: \quad \sum_{m=1}^{M} x_{m,t,n} \leq 1$$

3. **Class Correctness:** The final predicted class for an entire track must match the ground truth class. This is a crucial and non-trivial constraint. The class for a track m is determined by first averaging the logit vectors of all queries assigned to it, and then taking the argmax. Let $L_{t,n,c}$ be the logit for class c from query n at frame t. The average logit for track m and class c is $\bar{L}_{m,c} = \frac{1}{T}\sum_{t=1}^{T}\sum_{n=1}^{N} L_{t,n,c} \cdot x_{m,t,n}$. The constraint ensures that for a track m with ground truth class c_m^*, its average logit is highest for that class:
$$\forall m, \forall c' \neq c_m^* : \quad \bar{L}_{m,c_m^*} \geq \bar{L}_{m,c'} + \epsilon$$
where ϵ is a small margin.

This ILP formulation balances the trade-off between maximizing IoU and maintaining class consistency over the entire track, which a greedy algorithm cannot do. Solving it (in chunks for computational tractability) provides an empirical performance ceiling, revealing the substantial gap caused by the brittleness of frame-to-frame matching.

3.2 Qualitative Analysis with TrackLens

To complement our quantitative findings, we developed **TrackLens**, an interactive visualization tool for the qualitative analysis of tracking behavior. The tool allows us to step through videos frame by frame and inspect the full set of generated queries, their predicted masks, their class labels, and their similarity scores to queries in adjacent frames.

This provides direct visual evidence of the failure modes that our ILP oracle quantifies. For example, we can observe precisely when an important object query is incorrectly matched to a null query, simply because that assignment contributes to a slightly higher global similarity score for the Hungarian algorithm. It also helps identify scenarios where tracking fails due to temporary occlusions or multiple similar objects, motivating the need for a more robust, temporally-aware matching strategy. An overview of the TrackLens interface is shown in Figure 1.

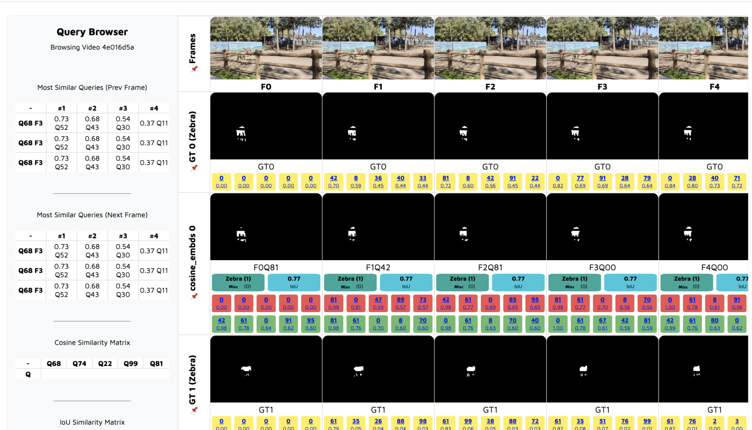

Figure 1: Overview of the query browser interface. Rows represent query tracks across time, columns show video frames. The main panel displays the query-frame matrix, while the side panel shows relationship metrics. Top row: original video frames; white backgrounds: ground truth masks; colored backgrounds: strategy-specific tracklets.

3.3 Robust Tracking via Multi-Frame Query Aggregation

Motivated by the performance gap identified by our oracle, we propose a simple yet effective modification to the MinVIS tracking algorithm. Instead of relying solely on the similarity between queries in two consecutive frames, our method aggregates information from a temporal window of past frames.

Aggregated Similarity. At each frame t, when matching the set of current queries Q^t to the established tracks, we look back at the query embeddings of the last W frames. The standard similarity matrix is replaced by an aggregated matrix, S^t_{agg}. This matrix is a weighted sum of similarities between the current queries and the queries from each frame in the temporal window:

$$S^t_{\text{agg}} = \sum_{i=1}^{W} w_i \cdot S^{t,t-i}$$

where $S^{t,t-i}$ is the cosine similarity matrix between queries at frame t and frame $t-i$, and w_i are weights that sum to one ($\sum w_i = 1$).

Weighting Scheme. While various weighting schemes can be employed, such as exponential decay ($w_i \propto \alpha^{i-1}$), our experiments show that a simple **uniform weighting** ($w_i = 1/W$) provides strong and consistent performance without the need for additional hyperparameter tuning. This suggests that the presence of a query in the recent temporal context, rather than its precise recency, is the most important signal.

Once S^t_{agg} is computed, the Hungarian algorithm is applied as in the original MinVIS to find the optimal assignment. This temporal smoothing makes the matching process more resilient to noisy similarity scores caused by occlusions, rapid motion, or distracting objects, thereby preventing tracking failures and reducing identity switches. This entire procedure is a lightweight, training-free post-processing step.

4 EXPERIMENTS

4.1 EXPERIMENTAL SETUP

Datasets. We conduct evaluations on three widely used VIS benchmarks: YouTube-VIS 2019 and 2021 (YTVIS) Yang et al. (2019), and Occluded VIS (OVIS) Qi et al. (2022). YTVIS-19 and YTVIS-21 contain 2,238/2,985 training and 302/421 validation videos, respectively, across 40 object categories. The OVIS dataset, with 607 training and 140 validation videos over 25 categories, is designed to test robustness in scenarios with heavy object occlusions.

Evaluation Metrics. Following standard VIS protocol, we report Average Precision (AP) and Average Recall (AR). AP is the primary metric, averaged over 10 Intersection-over-Union (IoU) thresholds from 0.50 to 0.95. AR measures the fraction of ground-truth instances successfully matched by at least one prediction, reported for the top-1 (AR1) and top-10 (AR10) predictions per frame.

Baselines. Our method, MinVIS-W, is a training-free enhancement built on top of MinVIS Huang et al. (2022). Therefore, our direct baseline is the official MinVIS framework, using the publicly available model with a ResNet-50 (R50) He et al. (2015) backbone pre-trained on COCO Lin et al. (2015).

Implementation Details. For our **MinVIS-Oracle**, the ILP solver requires a feasible solution that assigns a valid track to every ground truth object. Given imperfect frame-level predictions, such a solution may not always be possible. In these cases, we iteratively relax the ”Unique Assignment” constraint to cover fewer objects until a valid solution is found. This ensures the oracle produces a robust upper bound even with realistic model outputs.

Since the ground truth for the official OVIS validation set is not public, calculating our MinVIS-Oracle upper bound on it is impossible. Therefore, for all OVIS experiments, we partitioned the official training set, using 100 videos for our evaluation and the remainder for hyperparameter tuning. This split allows us to report oracle performance and fairly evaluate MinVIS-W on the same held-out data. For YTVIS, we report results on the official ‘val‘ sets and use the ‘train‘ set for tuning.

Our proposed **MinVIS-W** introduces one key hyperparameter: the window size, W. This was tuned on the designated hyperparameter split for each dataset. Based on this tuning, we use a simple uniform weighting scheme for aggregating scores. No modifications to the base model or additional training are required.

Table 1: Results on YouTube-VIS 2019 & 2021 and OVIS.

Dataset	Method	Backbone	AP	AP50	AP75	AR1	AR10
YouTube-VIS 2019	MinVIS	R50	47.74	70.94	50.96	45.87	55.18
	MinVIS-W (Ours)	R50	**50.95**	**73.97**	**55.74**	**47.88**	**57.95**
	MinVIS-Oracle	R50	*57.25*	*79.52*	*63.02*	*51.68*	*66.87*
YouTube-VIS 2021	MinVIS	R50	44.46	67.11	48.32	39.03	51.97
	MinVIS-W (Ours)	R50	**46.33**	**69.42**	**52.20**	**40.71**	**53.72**
	MinVIS-Oracle	R50	*52.39*	*73.62*	*60.34*	*42.76*	*64.33*
OVIS	MinVIS	R50	49.09	78.91	53.31	22.96	52.48
	MinVIS-W (Ours)	R50	**53.34**	**80.28**	**57.53**	**24.38**	**56.59**
	MinVIS-Oracle	R50	*69.24*	*93.94*	*81.86*	*28.99*	*73.66*

4.2 Main Results

Our results are summarized in Table 1. We first quantify the performance gap between the standard MinVIS and the theoretical maximum (MinVIS-Oracle), then demonstrate how MinVIS-W narrows this gap.

Oracle Performance Gap. The MinVIS-Oracle results reveal that tracking errors are a significant performance bottleneck. On YTVIS-19, the oracle achieves **57.25** AP, a full **9.51** points higher than the baseline MinVIS's **47.74** AP. This gap is even more pronounced on the challenging OVIS dataset, where the oracle reaches **69.24** AP compared to the baseline's **49.09**—a **20.15**-point difference attributable solely to tracking and class assignment errors. This analysis clarifies that a significant portion of the performance drop arises from tracking errors, with the remaining gap to perfect evaluation reflecting limitations in detection and segmentation quality.

MinVIS-W Performance. Our proposed MinVIS-W method narrows this performance gap using a simple, training-free approach. On YTVIS-19, it achieves an AP of **50.95**, recovering over **3.2** AP points and closing more than a third of the gap to the oracle. On YTVIS-21, it provides a solid **+1.87** AP boost. The improvement is even more significant on the occlusion-heavy OVIS dataset, where MinVIS-W improves performance to **53.34** AP (**+4.25**), demonstrating its utility in challenging scenarios. These results confirm that our multi-frame aggregation strategy provides a more robust matching signal, leading to more consistent tracking and higher overall performance.

5 Conclusion

In this work, we analyzed and enhanced the tracking component of MinVIS, a leading online Video Instance Segmentation (VIS) framework. Our investigation revealed that tracking errors constitute a significant bottleneck in VIS performance—substantially more than previously recognized. Through our oracle analysis, we demonstrated that on YTVIS-19, tracking errors alone account for a **9.51** AP performance deficit, while on the challenging OVIS dataset, this gap reaches **20.15** AP.

To facilitate comprehensive evaluation of tracking behaviors, we introduced TrackLens, an interactive visualization and analysis tool that enables detailed inspection of query relationships and tracklet formation across video frames. Using TrackLens, we identified a critical vulnerability in MinVIS's Hungarian matching approach: null queries frequently interfere with the matching of significant queries, causing the algorithm to prioritize global similarity maximization over meaningful instance associations.

To address this fundamental issue, we proposed MinVIS-W, a training-free enhancement that leverages multi-frame temporal context for more robust matching decisions. By aggregating similarity information across a temporal window, our method significantly reduces identity switches and recovers a substantial portion of the performance gap between baseline matching and optimal assignment.

Our experimental results demonstrate consistent improvements across standard benchmarks, with MinVIS-W achieving notable gains of **+3.21** AP on YTVIS-19 and **+1.87** AP on YTVIS-21, while maintaining robustness on the occlusion-heavy OVIS dataset. These improvements come at no additional training cost and require minimal computational overhead.

We believe that TrackLens, with its comprehensive visualization capabilities, will serve as a valuable resource for the VIS community to better understand and improve online tracking algorithms. Our work highlights that significant performance gains in video instance segmentation can be achieved by focusing on the often-overlooked tracking component, opening new avenues for future research in robust temporal association methods.

References

Nicolas Carion, Francisco Massa, Gabriel Synnaeve, Nicolas Usunier, Alexander Kirillov, and Sergey Zagoruyko. End-to-end object detection with transformers. In Andrea Vedaldi, Horst Bischof, Thomas Brox, and Jan-Michael Frahm (eds.), *Computer Vision – ECCV 2020*, pp. 213–229, Cham, 2020. Springer International Publishing. ISBN 978-3-030-58452-8.

Bowen Cheng, Anwesa Choudhuri, Ishan Misra, Alexander Kirillov, Rohit Girdhar, and Alexander G. Schwing. Mask2former for video instance segmentation, 2021. URL `https://arxiv.org/abs/2112.10764`.

Bowen Cheng, Ishan Misra, Alexander G Schwing, Alexander Kirillov, and Rohit Girdhar. Masked-attention mask transformer for universal image segmentation. In *Proceedings of the IEEE/CVF conference on computer vision and pattern recognition*, pp. 1290–1299, 2022.

Fei He, Haoyang Zhang, Naiyu Gao, Jian Jia, Yanhu Shan, Xin Zhao, and Kaiqi Huang. Inspro: Propagating instance query and proposal for online video instance segmentation. *Advances in Neural Information Processing Systems*, 35:19370–19383, 2022.

Kaiming He, Xiangyu Zhang, Shaoqing Ren, and Jian Sun. Deep residual learning for image recognition, 2015. URL `https://arxiv.org/abs/1512.03385`.

De-An Huang, Zhiding Yu, and Anima Anandkumar. Minvis: A minimal video instance segmentation framework without video-based training. *Advances in Neural Information Processing Systems*, 35:31265–31277, 2022.

Junlong Li, Bingyao Yu, Yongming Rao, Jie Zhou, and Jiwen Lu. Tcovis: Temporally consistent online video instance segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision*, pp. 1097–1107, 2023.

Tsung-Yi Lin, Michael Maire, Serge Belongie, Lubomir Bourdev, Ross Girshick, James Hays, Pietro Perona, Deva Ramanan, C. Lawrence Zitnick, and Piotr Dollár. Microsoft coco: Common objects in context, 2015. URL `https://arxiv.org/abs/1405.0312`.

Jiyang Qi, Yan Gao, Yao Hu, Xinggang Wang, Xiaoyu Liu, Xiang Bai, Serge Belongie, Alan Yuille, Philip H. S. Torr, and Song Bai. Occluded video instance segmentation: A benchmark, 2022. URL `https://arxiv.org/abs/2102.01558`.

Yuqing Wang, Zhaoliang Xu, Xinlong Wang, Chunhua Shen, Baoshan Cheng, Hao Shen, and Huaxia Xia. End-to-end video instance segmentation with transformers. In *Proc. IEEE Conf. Computer Vision and Pattern Recognition (CVPR)*, 2021.

Nicolai Wojke, Alex Bewley, and Dietrich Paulus. Simple online and realtime tracking with a deep association metric. In *2017 IEEE international conference on image processing (ICIP)*, pp. 3645–3649. IEEE, 2017.

Junfeng Wu, Yi Jiang, Wenqing Zhang, Xiang Bai, and Song Bai. Seqformer: a frustratingly simple model for video instance segmentation. *arXiv preprint arXiv:2112.08275*, 2021.

Junfeng Wu, Qihao Liu, Yi Jiang, Song Bai, Alan Yuille, and Xiang Bai. In defense of online models for video instance segmentation. In *European Conference on Computer Vision*, pp. 588–605. Springer, 2022.

Linjie Yang, Yuchen Fan, and Ning Xu. Video instance segmentation. In *Proceedings of the IEEE/CVF international conference on computer vision*, pp. 5188–5197, 2019.

Shusheng Yang, Yuxin Fang, Xinggang Wang, Yu Li, Chen Fang, Ying Shan, Bin Feng, and Wenyu Liu. Crossover learning for fast online video instance segmentation. In *Proceedings of the IEEE/CVF International Conference on Computer Vision (ICCV)*, pp. 8043–8052, October 2021.

A Appendix

A.1 Visualization of Tracking

To provide a clearer understanding of the challenges in query-based tracking and the motivation for our multi-frame aggregation method, this section visualizes several key failure modes of the standard MinVIS framework.

The Problem of Null Queries. Query-based models like Mask2Former generate a fixed, large number of queries per frame, most of which do not correspond to any actual object and are considered ”null” queries. The Hungarian algorithm, used for matching, seeks a globally optimal assignment for all queries, including these null ones. This can lead to situations where a correct match for a real object is sacrificed to achieve a slightly better overall similarity score across all queries. Figure 4 demonstrates this issue, showing a sample frame alongside the complete set of query masks generated for it, the majority of which are empty.

Figure 2: Most queries are null. (Left is a sample frame, right is the list of queries masks for that frame)

Imperfectly Discriminative Queries. The success of query matching depends on the assumption that queries for the same object instance remain highly similar across frames, while queries for different objects are distinct. However, in practice, these query embeddings are not perfectly discriminative. The resulting similarity matrix is often noisy, with high similarity scores appearing between queries of different objects, or low scores for the same object during challenging scenarios like occlusion or rapid motion. Figure 3 shows a visualization of a similarity matrix, where the noise and ambiguity are evident.

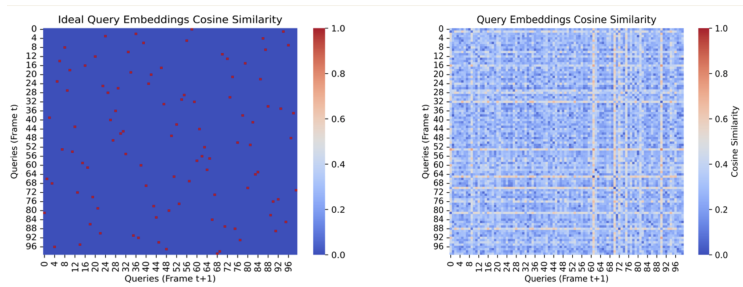

Figure 3: Queries are not perfectly discriminative: The similarity matrix between queries is noisy.

The Fragility of Frame-to-Frame Matching. The core tracking mechanism in MinVIS relies on matching query embeddings between consecutive frames. While generally effective, this approach can be brittle. A single incorrect match can cause a tracking sequence to *derail*, where the identity of a track is permanently lost or swapped. Figure A.1 illustrates such a scenario. Once an object's track is associated with an incorrect query (often a null query), the error propagates through subsequent frames, as the system tries to match the new, incorrect query forward.

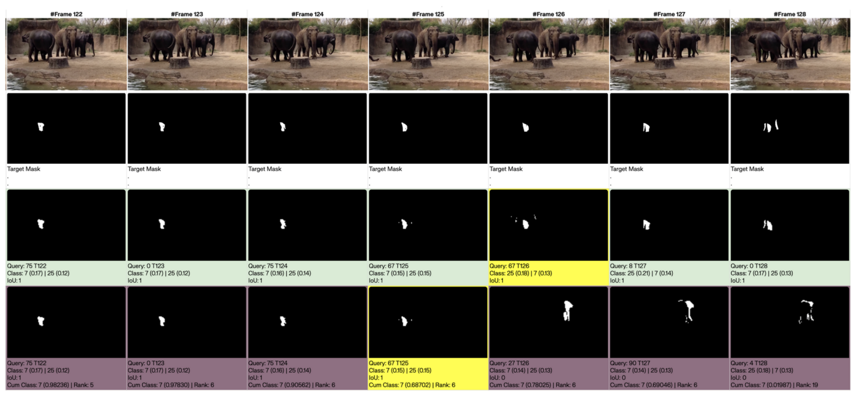

Figure 4: A tracking failure in MinVIS. From top to bottom: (1) original video frames, (2) ground truth masks for an object, (3) the correct track from our oracle (green), and (4) the derailed track from MinVIS's greedy matching (red). A single mistake causes the track to fail.

A.2 Ablations

We perform an ablation study to analyze the impact of the temporal window size W and different weighting schemes on the OVIS dataset, with results shown in Figure 5. The analysis reveals that a small window size of $W = 2$ yields the most substantial performance gains in both AP and AR1. Larger window sizes lead to a decline in performance, suggesting that outdated temporal information can introduce noise and degrade matching quality. Among the weighting strategies, a simple uniform weighting consistently delivers the strongest results, indicating that the presence of recent historical context is more critical than its precise recency.

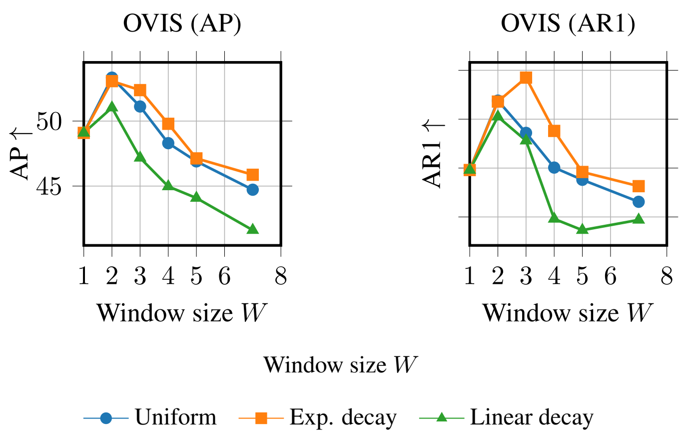

Figure 5: Effect of temporal window size W and weighting scheme on AP and AR1 for OVIS.